One-Way Within-Subjects ANOVA Demo

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One-way Within-Subjects ANOVA | Research Context Prompt

A local stress researcher is interested in the relationship between psychosocial stress levels and processing speed. After conducting a brief literature review they have hypothesized that regulating feelings of stress impairs concurrent performance of processing speed.

The researcher conducted a study where psychosocial stress was manipulated into three conditions: low stress (**Calm**), mild stress (**Mild**), and strong stress (**Strong**). Money was tight and recruitment was difficult, so they implemented a within-subjects, repeated measures design. Data was collected from 20 participants that consented to participate. The dependent measure in this task was processing speed ability measured in 10-second intervals (i.e., 1.3 = 13 seconds!!)

One-way Within-Subjects ANOVA | Data

```
library(psych)
library(tidyverse)
library(jmv)
library(ggpubr)
library(ggrain)
library(apaTables)
library(ez)
library(rstatix)

dat_stress <- read.csv("RM.ANOVA.Demo.csv")
dim(dat_stress)
```

```
## [1] 20  4
```

```
head(dat_stress)
```

```
##   ID Calm Mild Strong
## 1  1  3.4  5.0    5.9
## 2  2  3.9  4.0    5.4
## 3  3  4.1  4.4    4.7
## 4  4  3.7  4.3    5.9
## 5  5  4.0  5.0    5.1
## 6  6  4.6  4.1    6.4
```

One-way Within-Subjects ANOVA | Descriptive Stats

```
# Descriptive Stats
describe(dat_stress[-1])
```

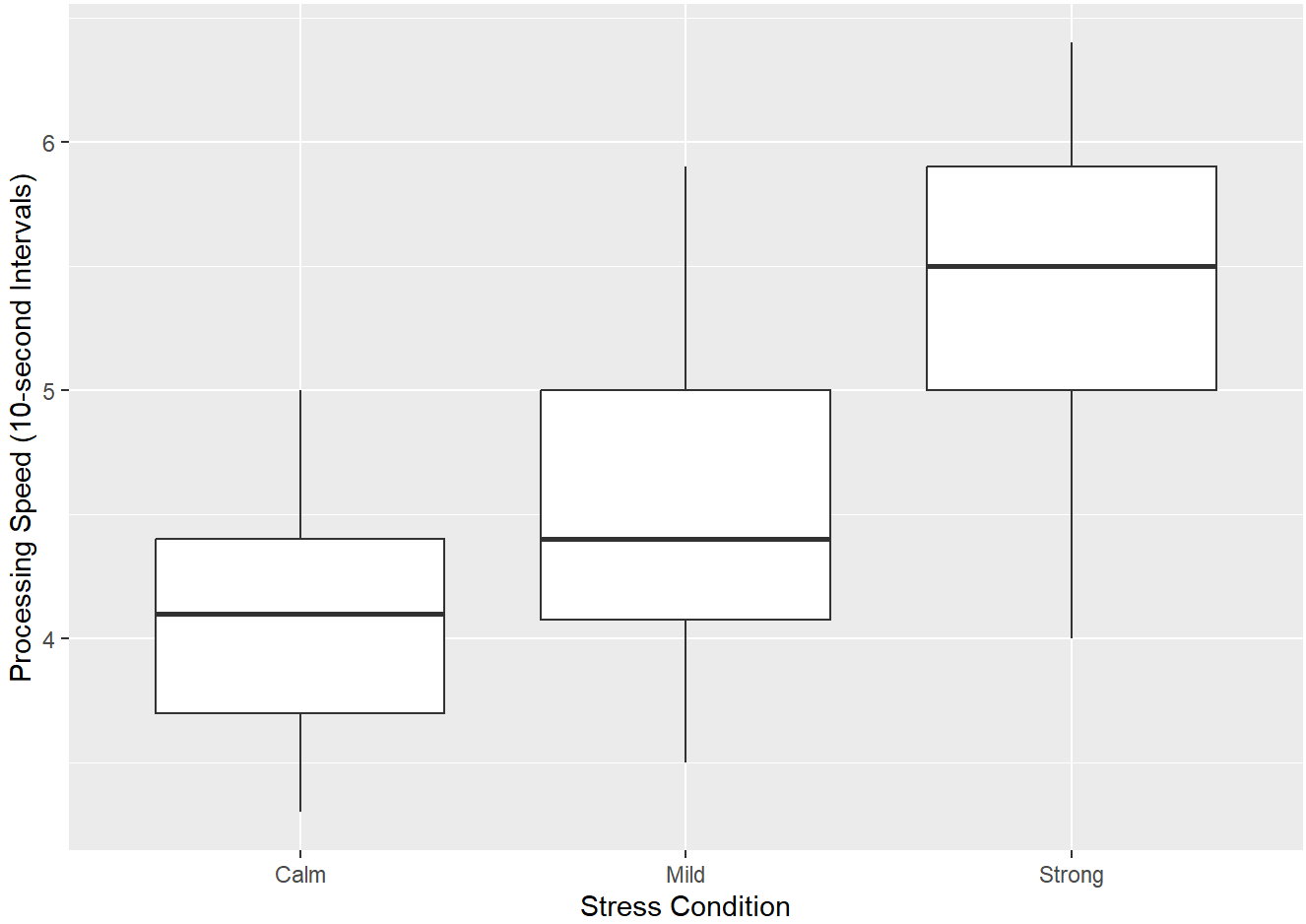
```
##        vars  n mean   sd median trimmed  mad min max range  skew kurtosis   se
## Calm      1 20 4.06 0.45    4.1    4.04 0.59 3.3 5.0   1.7  0.20    -0.88 0.10
## Mild      2 20 4.57 0.76    4.4    4.53 0.82 3.5 5.9   2.4  0.53    -1.06 0.17
## Strong    3 20 5.43 0.67    5.5    5.47 0.59 4.0 6.4   2.4 -0.50    -0.87 0.15
```

One-way Within-Subjects ANOVA | Visualizations

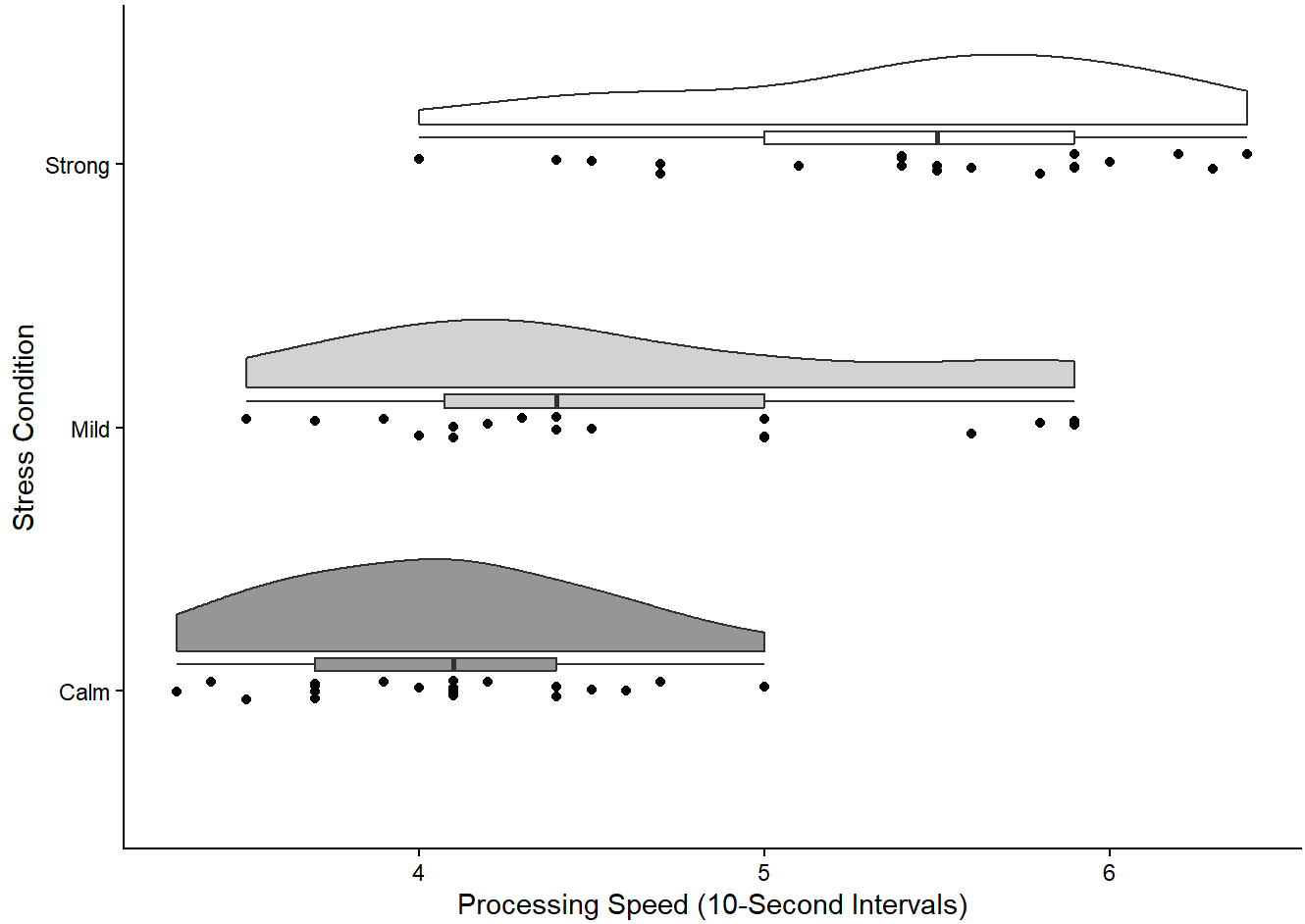
First we need to make the wide data into a long format

```
# Create Long Data Frame using tidyr::gather()

dat_stressL<- dat_stress %>%
  gather(key="Stress_Condition",
         value="Speed",
         Calm,
         Mild,
         Strong,
         factor_key = TRUE)

dat_stressL$ID <- factor(dat_stressL$ID)

# OR you can use the tidyr::pivot_longer() function workflow

dat_stressL2 <- dat_stress %>%
  pivot_longer(cols = -ID,
               names_to = "Stress_Condition",
               values_to = "Speed")

dat_stressL2$Stress_Condition<-factor(dat_stressL2$Stress_Condition,labels = c("Calm","Mild","St
rong"))

dat_stressL2$ID <- factor(dat_stressL2$ID)
```

```
# Boxplot Visualization
ggplot(data = dat_stressL,
       mapping = aes(y = Speed, x = Stress_Condition)) +
  geom_boxplot() +
  ylab("Processing Speed (10-second Intervals)") +
  xlab("Stress Condition")
```




```
# Raincloud Plot Visualization
ggplot(data = dat_stressL,
       aes(x = Stress_Condition, y = Speed,
           fill = Stress_Condition))+
  geom_rain(show.legend=FALSE) +
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Processing Speed (10-Second Intervals)")+
  xlab("Stress Condition")+
  theme_classic()
```



One-way Within-Subjects ANOVA | Sphericity Violation Check

```
# JMV Code does NOT require Long data frame object

anovaRM(data=dat_stress,
        rm = list(
          list(label = 'Stress Condition',
               levels = list('Calm','Mild','Strong'))),
        rmCells = list(
          list(measure = 'Calm', cell = 'Calm'),
          list(measure = 'Mild', cell = 'Mild'),
          list(measure = 'Strong', cell = 'Strong')),
        rmTerms= list('Stress Condition'),
        spherTests = TRUE)
```

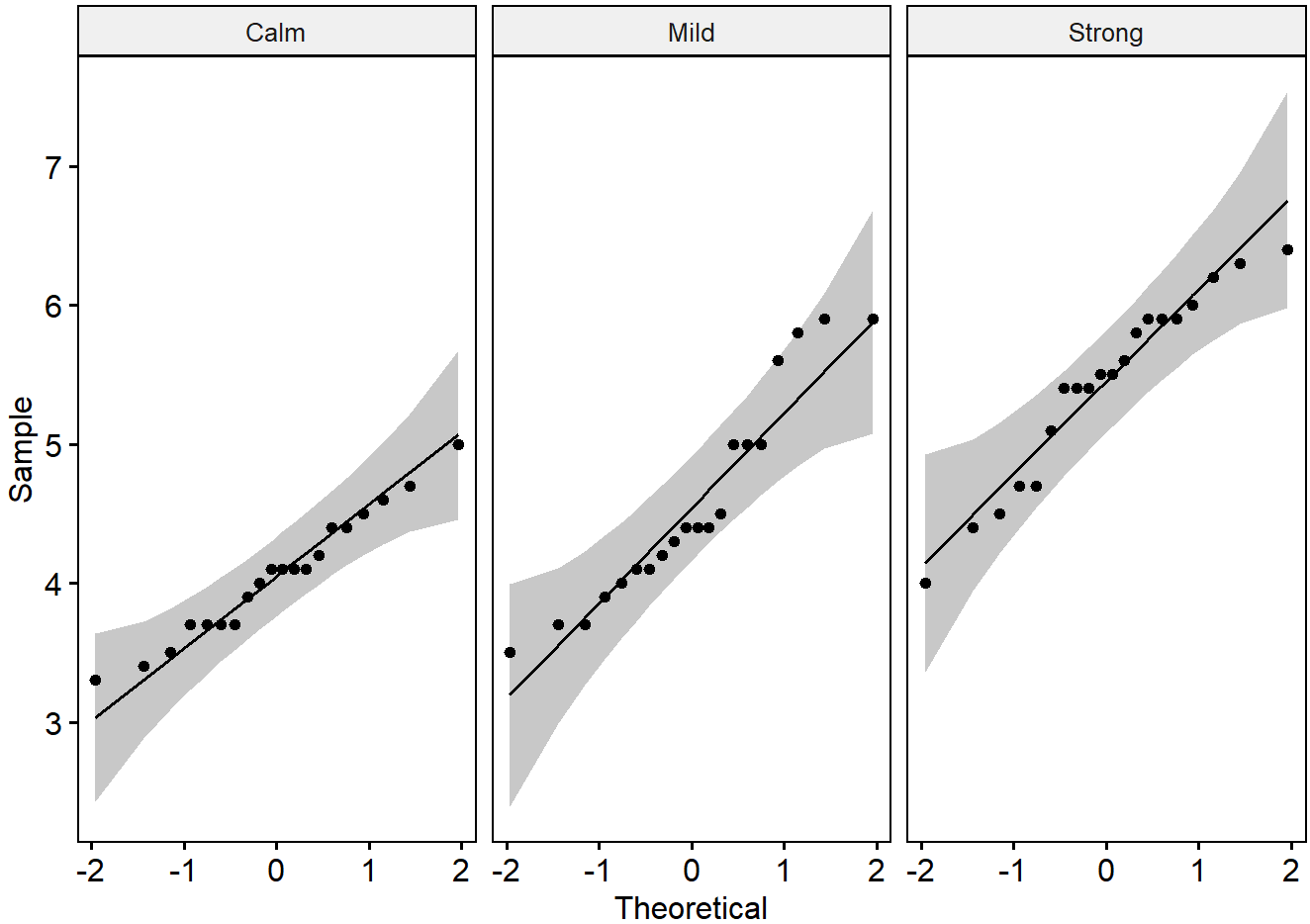
```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ─────────────────────────────────────────────────────────────────────────────────────
##                        Sum of Squares    df    Mean Square    F           p
##  ─────────────────────────────────────────────────────────────────────────────────────
##    Stress Condition          19.30300     2      9.6515000    22.22276    0.0000004
##    Residual                  16.50367    38      0.4343070
##  ─────────────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  Between Subjects Effects
##  ─────────────────────────────────────────────────────────────────────
##                Sum of Squares    df    Mean Square    F    p
##  ─────────────────────────────────────────────────────────────────────
##    Residual          6.849833    19      0.3605175
##  ─────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  ASSUMPTIONS
##
##  Tests of Sphericity
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##                        Mauchly's W    p            Greenhouse-Geisser ε    Huynh-Feldt ε
##  ─────────────────────────────────────────────────────────────────────────────────────────────
##    Stress Condition      0.9655453    0.7293799               0.9666929         1.000000
##  ─────────────────────────────────────────────────────────────────────────────────────────────
```

One-way Within-Subjects ANOVA | Normality Assumption Check

```
# ggpubr::ggqqplot() requires long data frame

dat_stressL %>%
  group_by(Stress_Condition) %>%
  shapiro_test(Speed)
```

```
## # A tibble: 3 × 4
##   Stress_Condition variable statistic      p
##   <fct>            <chr>        <dbl>  <dbl>
## 1 Calm             Speed        0.974 0.841
## 2 Mild             Speed        0.910 0.0632
## 3 Strong           Speed        0.944 0.286
```

```
ggqqplot(dat_stressL,
         'Speed',
         facet.by = "Stress_Condition")
```



One-way Within-Subjects ANOVA | Conducting the ANOVA

```
options(digits = 3)
anovaRM(data=dat_stress,
        rm = list(
          list(label = 'Stress',levels = list('Calm','Mild','Strong'))),
        rmCells = list(
          list(measure = 'Calm', cell = 'Calm'),
          list(measure = 'Mild', cell = 'Mild'),
          list(measure = 'Strong', cell = 'Strong')),
        rmTerms= list('Stress'),
        effectSize = 'eta',
        postHoc='Stress',
        postHocCorr = list("bonf"),
        emMeans=list(
          list('Stress')),
        emmTables=TRUE,
        emmPlots=FALSE,
        spherTests = TRUE,
        spherCorr = list('none', 'GG'))
```

```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ──────────────────────────────────────────────────────────────────────────────────────────────
##                Sphericity Correction    Sum of Squares    df       Mean Square    F       p
η²
##  ──────────────────────────────────────────────────────────────────────────────────────────────
##    Stress      None                              19.3        2          9.651    22.2    < .
001    0.453
##                Greenhouse-Geisser                19.3     1.93          9.984    22.2    < .
001    0.453
##
##    Residual    None                              16.5       38          0.434
##                Greenhouse-Geisser                16.5    36.73          0.449
##  ──────────────────────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  Between Subjects Effects
##  ──────────────────────────────────────────────────────────────────────
##                Sum of Squares    df    Mean Square    F    p        η²
##  ──────────────────────────────────────────────────────────────────────
##    Residual              6.85    19          0.361
##  ──────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  ASSUMPTIONS
##
##  Tests of Sphericity
##  ────────────────────────────────────────────────────────────────────────
##              Mauchly's W    p        Greenhouse-Geisser ε    Huynh-Feldt ε
##  ────────────────────────────────────────────────────────────────────────
##    Stress          0.966    0.729                   0.967             1.00
##  ────────────────────────────────────────────────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - Stress
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Stress         Stress    Mean Difference    SE       df      t        p-bonferroni
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Calm      -    Mild               -0.515    0.211    19.0    -2.44           0.074
##              -    Strong             -1.375    0.190    19.0    -7.25    < .001
##    Mild      -    Strong             -0.860    0.223    19.0    -3.86           0.003
##  ──────────────────────────────────────────────────────────────────────────────────────────
##
```

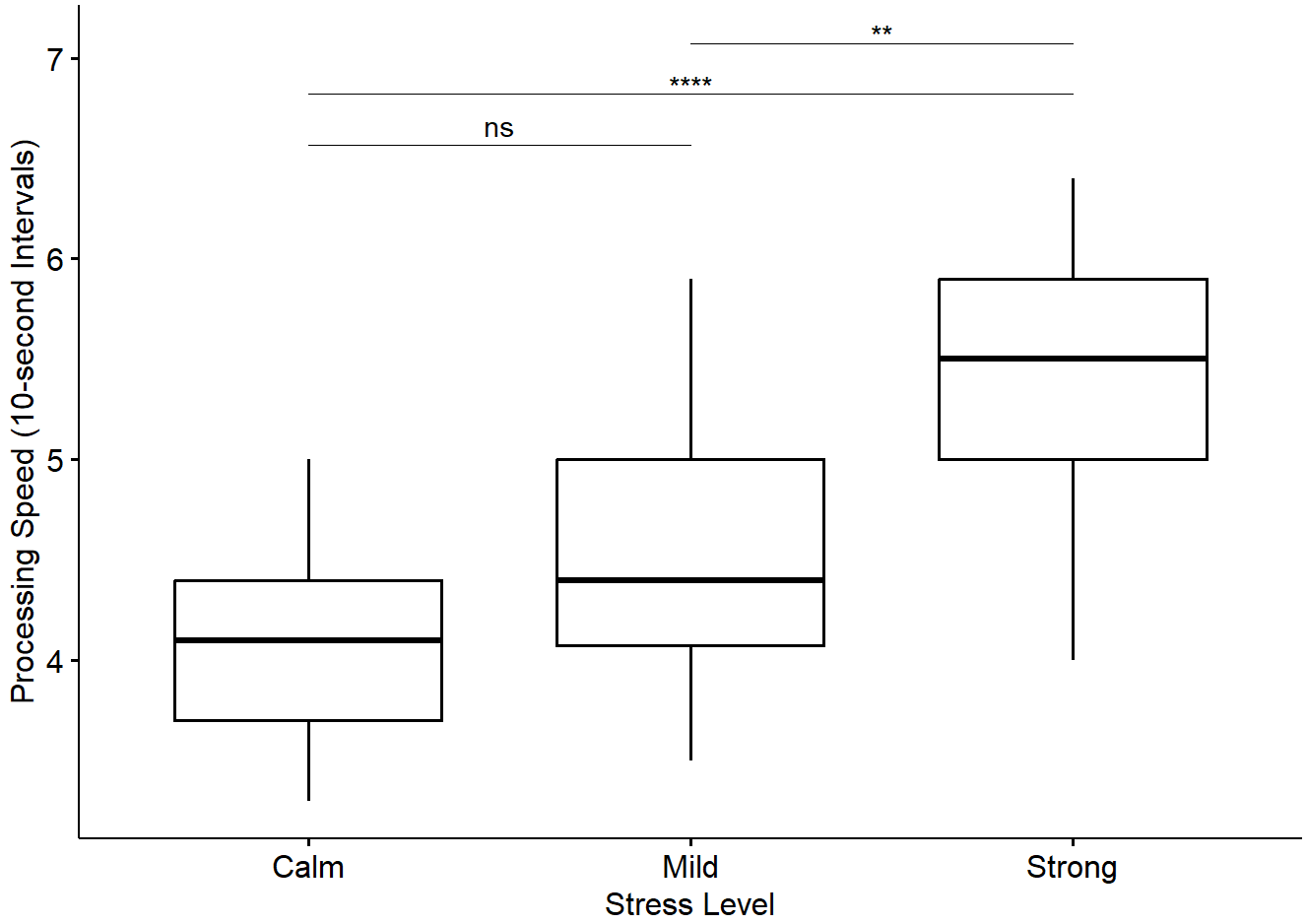
```
##
##  ESTIMATED MARGINAL MEANS
##
##  STRESS
##
##  Estimated Marginal Means - Stress
##  ──────────────────────────────────────────────
##    Stress    Mean    SE       Lower    Upper
##  ──────────────────────────────────────────────
##    Calm      4.06    0.101     3.84     4.27
##    Mild      4.57    0.169     4.22     4.92
##    Strong    5.43    0.150     5.12     5.74
##  ──────────────────────────────────────────────
```

One-way Within-Subjects ANOVA | ANOVA Output: Manual Post-Hoc Tests Code

```
pwc <- dat_stressL %>%
  pairwise_t_test(Speed~Stress_Condition,
                  paired=TRUE,
                  p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "Stress_Condition")
```

One-way Within-Subjects ANOVA | Professional ANOVA Visualization Code

```
ggboxplot(dat_stressL,
          x = "Stress_Condition",
          y = "Speed") +
  stat_pvalue_manual(pwc,
                     #hide.ns = TRUE,
                     tip.length = 0) +
  labs(y = "Processing Speed (10-second Intervals)", x = "Stress Level")
```



One-way Within-Subjects ANOVA | APA Style ANOVA Tables

```
model <- ezANOVA(dat_stressL,
                 dv=.(Speed),
                 wid=.(ID),
                 within=.(Stress_Condition),
                 type = 3,
                 detailed=TRUE)

apa.ezANOVA.table(model,
                  correction="none",
                  table.title = "Repeated Measure ANOVA",
                  table.number=3)
```

```
##
##
## Table 3
##
## Repeated Measure ANOVA
##
##         Predictor df_num df_den  SS_num SS_den       F    p ges
##       (Intercept)      1     19 1316.95   6.85 3652.95 .000 .98
##  Stress_Condition      2     38   19.30  16.50   22.22 .000 .45
##
## Note. df_num indicates degrees of freedom numerator. df_den indicates degrees of freedom deno
minator.
## p-values based on assumed sphericity.
## p-values and degrees of freedom in the table incorporate this correction.
## SS_num indicates sum of squares numerator. SS_den indicates sum of squares denominator.
## ges indicates generalized eta-squared.
##
```